

GAMIKA JAYAWARDHANA

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SUMMARY

Ambitious fourth-year Electronics and Computer Science undergraduate (GPA 3.6/4.0) specialising in Artificial Intelligence and Machine Learning. Proven ability to bridge the gap between theoretical deep learning and production-ready software, demonstrated through ongoing research in edge-AI quantisation and the deployment of stateful, multi-agent LLM systems. Combines advanced computer vision expertise with full-stack engineering proficiency to take complex models from initial notebooks to fully deployed, scalable products.

TECHNICAL SKILLS

Artificial Intelligence (AI): LLM APIs, RAG, NLP, LangChain, LangGraph, Multi-Agent Systems, prompt engineering, vector databases (MongoDB Vector Search, Pinecone, ChromaDB), Generative AI, Computer Vision, n8n Automation

Machine Learning (ML): PyTorch, TensorFlow/Keras, scikit-learn, pandas, NumPy, OpenCV, MediaPipe, ONNX, TensorRT, INT8/FP16 Quantisation, Weight Pruning, knowledge distillation, Deep Learning, Transformers

Languages and Frameworks: Python (FastAPI), Java (SpringBoot), TypeScript/JavaScript (React, Next.js), SQL, C

Deployment & MLOps: FastAPI, Docker, GitHub Actions (CI/CD), AWS, Git/GitHub, Linux, REST APIs, WebSocket

Foundations: Data Structures & Algorithms, statistics & probability, linear algebra, data preprocessing

EDUCATION

BSc (Hons) in Electronics and Computer Science — University of Kelaniya, Sri Lanka Jul 2023 – Present (Expected 2027)
Double major in Electronics and Computer Science. Current GPA: 3.6 / 4.0.

Relevant coursework: Data Structures & Algorithms, Statistics & Probability, Linear Algebra, Database Systems, Machine Learning, Artificial Intelligence

Diploma in English — Sabaragamuwa University of Sri Lanka Jan 2021 – Dec 2021

RESEARCH EXPERIENCE

Lightweight Monocular Depth Estimation for Assistive Navigation on Edge Devices In progress

Python, PyTorch, ONNX, TensorRT, Vision Transformers, Knowledge Distillation, Quantisation, OpenCV

- Designed a real-time depth-estimation pipeline using knowledge distillation from a DINOv2 teacher (98%+ $\delta < 1.25$) to a MobileNetV4 student that retains 93%+ accuracy at a fraction of the compute.
- Optimised models for edge deployment via ONNX export, INT8/FP16 quantisation, and weight pruning, targeting real-time inference on Raspberry Pi, NVIDIA Jetson, and smartphones.
- Designing a multi-model assistive navigation system for blind and visually impaired users, fusing depth estimation and obstacle detection into real-time audio/haptic feedback.

PROJECTS

MedOpsAI — A real-time event-driven control centre for hospital operations | [GitHub](#)

Python, FastAPI, WebSockets, LangGraph, LangChain, Pydantic, Next.js, TypeScript, RAG, Embedding Models, Pinecone

- Architected a stateful multi-agent hospital operations suite using LangGraph and FastAPI, implementing durable execution and PostgreSQL checkpointing to allow human-in-the-loop approvals for resource allocation.
- Engineered an advanced Hybrid Search RAG pipeline with Pinecone and HuggingFace models, combining dense and sparse embeddings with cross-encoder re-ranking to retrieve operational SOPs with sub-second latency.
- Integrated Llama 3.3 via Groq for ultra-fast agentic reasoning, structuring outputs into strict JSON schemas to populate a real-time, WebSocket-driven Next.js dashboard optimised for Core Web Vitals

Predictive Diagnostic ML Application for Diabetes Risk | [GitHub](#)

Python, scikit-learn, pandas, FastAPI, React, Docker, AWS, GitHub Actions

- Trained, evaluated, and serialised classification models for diabetes risk assessment, served through a FastAPI REST API with a React front end.
- Containerised with Docker and built a GitHub Actions CI/CD pipeline deploying automatically to AWS - a complete model-to-production workflow.

Conversational NLP Expense-Tracking WhatsApp Bot | [GitHub](#)

Python, FastAPI, Groq API (Llama 3.3 70B), LangChain, MongoDB, Meta WhatsApp Cloud API, Webhook, Pydantic

- Built and deployed a WhatsApp chatbot that parses natural-language spending messages through webhook endpoints, using LangChain with Llama 3.3 70B and Pydantic-validated structured outputs (retry-on-failure) to extract and auto-categorise expenses into MongoDB.

AI Email Assistant — Chrome Extension + LLM Backend | [GitHub](#)

Java, Spring Boot, JavaScript, Manifest V3, LLM APIs, RAG, MongoDB Vector Search

- Architected a retrieval-augmented generation pipeline over MongoDB Vector Search feeding a Spring Boot LLM backend, exposed through a Manifest V3 Chrome extension for in-browser email summarisation and reply drafting.

HarvestLink — Microservices Platform with ML Price Forecasting (Group Project) | [GitHub](#)

Java, Spring Boot, Apache Kafka, React, MongoDB, Python, scikit-learn

- Collaborated in a team to build an event-driven Spring Boot microservices agri-commerce platform with a Kafka event pipeline and React storefront.
- Developed the Python crop-price forecasting engine, training scikit-learn models on 130,000+ rows of historical climate and price data.

WORK EXPERIENCE

Freelance Developer — Greams Oct 2023 – Present

- Engineered workflow-automation and AI-integration systems for clients using n8n, LLM APIs, and Python, reducing manual processing time by an estimated 30-40% per client.
- Delivered 10+ full-stack web and automation systems end-to-end (React/Next.js, FastAPI, Spring Boot), deployed and maintained in production on Vercel, Netlify, and cPanel hosting.
- Completed 50+ additional client engagements in UI/UX and design, managing scope, deadlines, and communication directly with clients.

Trainee, Digital Products Promoting Assistant — Bank of Ceylon May 2022 – Feb 2023

- Supported digital-banking customer onboarding and collaborated with technical teams on service and user-experience improvements.

ADDITIONAL INFORMATION

Languages: English, Sinhala

Certifications: MySQL Implementation Certified Associate (Oracle) • Introduction to Linux – LFS101 (Linux Foundation) • FastAPI & Backend Development Fundamentals (Packt)

Leadership and Volunteering: Senior Editor – VidE Club, University of Kelaniya • Head of Designs – KelaniSTEAM • External Relations Coordinator – Poverty & Hunger Avenue, Leo Club

REFERENCES

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